

Yuanlong WEI

Tel: +8615820000219 | Email: yusonw@outlook.com

Dezhou City, Shandong Province, P.R.China

EDUCATION

Harbin Institute of Technology (C9 / 985 / Double First-Class)

Harbin, Heilongjiang, China

Master of Engineering in Traffic and Transportation Engineering

Sep. 2022 – Jun. 2025

Thesis: *Research on Intelligent Passivity-Based Control Techniques for Connected and Automated Vehicle Platoons*

Jilin University (985 / Double First-Class)

Changchun, Jilin, China

Bachelor of Engineering in Traffic Engineering

Sep. 2018 – Jun. 2022

GPA: 3.62/4.0 (89.14/100), Ranked 7/66 (Top 11%)

Thesis: *Traffic Organization Optimization Design at Xinxing South Road – Xiushi Street Intersection*

RESEARCH EXPERIENCES

National Natural Science Foundation – Cooperative Control for Vehicle Platoon Systems under Coupled Conditions of Switching Communication Topology and Unfavorable Road Alignment

Project No.: 62303134

Jan. 2024 – Dec. 2026

Role: Key Project Participant / Supervisor: Associate Professor Yunfei Yin

- Published two papers under this grant: *V2V-Based Cooperative Control of Heterogeneous CAV Platoons: An Intelligent VO-IDA Approach (highest quality to date)* and *Model-Free Sliding Mode Control for Path Following in Autonomous Vehicle*.
- Highly involved in grant proposal writing, including research significance, literature review, research content, and methodology. Contributed to drafting and language polishing across 14 revisions.
- Outcome: Strengthened competence in high-quality academic writing with rigorous logic and precise language.

HIT “0-1 Original Exploration” Idea Fund Project – Multi-Vehicle Platoon Cooperative Control under Vehicle-Road Interference and Switching Communication Topology

Project No.: HIT.DZJJ.2023084

Nov. 2023 – Nov. 2024

Role: Project Leader / Supervisor: Associate Professor Yunfei Yin

- Published a paper under this grant: *V2V-Based Cooperative Control of Heterogeneous CAV Platoons: An Intelligent VO-IDA Approach*.
- Proposed a new intelligent control approach named "VO-IDA" for general control problems with chained integral features, which covers the cooperative control of CAV Platoons. Led formula derivation (600+ pages, 4 months).
- Outcome: Gained extensive academic experience in LaTeX-based writing and Inkscape-based exquisite vector illustration design.

PUBLICATIONS

Journal Papers

1. Yin, Y. (Supervisor), **Wei, Y. (1st co-author)**, Dong, Z., Xue, M., Vazquez, S., Wu, L.

V2V-Based Cooperative Control of Heterogeneous CAV Platoons: An Intelligent VO-IDA Approach

IEEE Internet of Things Journal, **2024**, 11(22): 36257.

ISSN: 2327-4662 | Status: Published

2. **Wei, Y. (1st author)**, Yin, Y., Jiang, W., Dong, Z., Vazquez, S., Xu, Z.

Model-Free NN Adaptive VO-IDA Control Against Uncertain Dynamics for CAV Platoons

IEEE Transactions on Industrial Informatics

Status: Under Review

3. Yin, Y., Xu, Z., **Wei, Y. (3rd author)**, Jiang, W., Yao, D., Dong, Z.

Model-Free Sliding Mode Control for Path Following in Autonomous Vehicle

Journal of the Franklin Institute, **2025**, 362(12): 107806.

ISSN: 1879-2693 | Status: Published

4. Yin, Y., Jiang, W., **Wei, Y. (3rd author)**, Shangguan, L., Xu, Z., Zong, Z.

Fusion Estimation of Tire Road Adhesion Coefficient Based on Tire-Icy Road Interaction Simulation and

Conference Papers

5. Xu, Z., Yin, Y., **Wei, Y. (3rd author)**, Jiang, W.

Improved Model Free Adaptive Control and Line-of-Sight Guidance for Autonomous Vehicle Path Following
IEEE International Conference on Industrial Technology (ICIT), **2025**.

6. Zhu, W., Yin, Y., Xu, Z., **Wei, Y. (4th author)**, Zeng, S.

Multi-objective Cooperative Control of Vehicle Platoon Based on Fuzzy MPC
IEEE International Conference on Industrial Technology (ICIT), **2025**.

ACADEMIC SKILLS

Algorithms, Theories & Implementation

- Learning Algorithms: Deep Learning, Reinforcement Learning, Genetic Algorithms
- Control Algorithms & Theories: Passivity-Based Control (IDA-PBC), Sliding Mode Control (SMC), Adaptive Control (Neural Network-Based), Model Predictive Control (MPC), Backstepping, Fuzzy Control (FC), Active Disturbance Rejection Control (ADRC), Linear System Theory, Lyapunov Stability Analysis
- Implementation: PyTorch + Python, MATLAB + Simulink (Proficient)

Programming Languages

- Python (Competent), C (Competent), MATLAB (Proficient), Excel VBA (Proficient)

General Academic Tools

- Literature Management: Zotero, Bookends (BibTeX integrating with LaTeX)
- Presentation: PowerPoint (Academic slide design)
- Graphics Tools: Inkscape (Vector graphics), MATLAB Data Visualization, Photoshop (Image editing), MindNode (Mind map), AutoCAD (Engineering drawing)
- Academic Writing: LaTeX (Proficient), MS Word (Proficient), Markdown (Competent)
- Data Processing: MATLAB Data Analysis (Proficient), Excel (Advanced functions)

Others

- Vehicle Dynamics Simulation (Proficient): CarSim-Simulink Co-Simulation, Powertrain Structures
- Intelligent Transportation Systems: Connected and Autonomous Vehicles, V2V Communication, Platoon Control, Operations Research, Traffic Management & Control, Traffic Flow Theory
- Traffic simulation: VISSIM (Proficient), TransCAD

Language Proficiency

- English (Fluent), Chinese (Native)

AWARDS & SCHOLARSHIPS

Second-Class Academic Scholarship, Harbin Institute of Technology. Twice (Nov. 2023, Nov. 2024)

Top-Class Academic Scholarship, Harbin Institute of Technology (Apr. 2023)

Outstanding Undergraduate Thesis, Jilin University (Jun. 2022)

Safe China Tour · Road Safety Innovation Competition, GAC Honda Automobile Co., Ltd. Third Prize, National Level (Dec. 2020)

College Outstanding Student, Jilin University. 4 times (2019-2022)

Second-Class Scholarship, Jilin University. 4 times (2019-2022)

Social Work Award, Jilin University (Apr. 2019)